PERCENTAGES

Short Revision Notes | Analyze Level — Break Information into Parts & Identify Relationships

1. Breaking Down the Percentage Concept

Component	What It Represents	Example: 25%
Percent Sign (%)	"Out of 100"	25 out of 100
Numerator Value	The part being considered	25 parts
Denominator Value	The whole (always 100)	100 parts
Decimal Equivalent	Value divided by 100	0.25
Fraction Equivalent	Part over whole	1/4

```
25% = 25/100 = 0.25 = 1/4

If percentage increases → value increases
If percentage decreases → value decreases
If percentage = 100% → whole = 1
If percentage > 100% → value > whole
If percentage < 100% → value < whole
```

2. Breaking Down Profit and Loss

Component	Profit	Loss
Definition	SP > CP	SP < CP
Formula	SP – CP	CP – SP
Percentage Base	Always CP	Always CP
Result	Positive return	Negative return
SP Formula	CP × (100 + P%)/100	CP × (100 – L%)/100

CP	SP	Profit	Profit %	Relationship
100	110	10	10%	SP = CP + 10% of CP
100	120	20	20%	SP = CP + 20% of CP
100	90	-10	10% Loss	SP = CP – 10% of CP
100	80	-20	20% Loss	SP = CP – 20% of CP

■ *Analysis:*
Profit/Loss always depends on Cost Price!
Profit% = (Profit ÷ CP) × 100   Loss% = (Loss ÷ CP) × 100
Profit% and Loss% are NOT calculated on Selling Price!

3. Breaking Down the Components of a Transaction

```
Cost Price (CP) → Marked Price (MP) → Selling Price (SP)
    What seller       What seller       What customer
```

```
    paid for item   marks as price   actually pays
```

Component	Meaning	Based On	Example
CP	What seller paid	Actual cost	Rs 800
MP	CP + Expected Profit	CP	Rs 1000
Discount	Reduction from MP	MP	Rs 100
SP	MP – Discount or CP ± P/L	MP or CP	Rs 900

```
If CP = 100, Markup = 20%, Discount = 10%
MP = 100 + 20% of 100 = 120
Discount = 10% of 120 = 12
SP = 120 - 12 = 108
Final Profit = 108 - 100 = 8% (not 10%!)
```

■ *Analysis:*
Why the analysis matters: Markup and Discount are based on DIFFERENT amounts.
Markup is based on CP; Discount is based on MP.
You cannot just add or subtract the percentages directly!

4. Comparing Profitability: Profit % vs Actual Profit

Case Study

Seller	CP	SP	Profit	Profit %
Ramesh	800	900	100	12.5%
Suresh	2500	2600	100	4%

■ *Analysis:*
Both earned the same profit (Rs 100), but Ramesh earned 12.5% while Suresh earned 4%.
Ramesh's transaction is MORE PROFITABLE.
Why? Ramesh earned Rs 100 on Rs 800 invested; Suresh earned Rs 100 on Rs 2500 invested.
Ramesh got a better return on investment.

```
Profit percentage shows efficiency, not just amount.
Higher profit% = Better investment return.
```

5. Analyzing Multi-Step Profit/Loss Problems

Scenario: Two-Step Profit

Step	Transaction	CP	SP	Profit %
1	Carpenter → Vendor	1800	2160	20%
2	Vendor → Customer	2160	2592	20%
Total	Carpenter → Customer	1800	2592	44%

■ *Analysis:*
Profit% is NOT additive! 20% + 20% ≠ 40%.
The second profit is calculated on a NEW CP (2160), not the original 1800.

```
Why?
Vendor's CP = 2160 (not 1800)
Vendor's Profit = 20% of 2160 = 432
Total Profit = 360 + 432 = 792
Total Profit% = 792/1800 × 100 = 44%

Method A: 20% + 20% = 40% (WRONG!)
Method B: Calculate step by step = 44% (CORRECT!)
Because percentages are calculated on different bases!
```

6. Analyzing the Relationship Between Markup and Discount

Markup %	Discount %	Final Profit/Loss	Analysis
20%	10%	8% Profit	Discount < Markup → Profit
20%	20%	4% Loss	Discount = Markup → Loss!
20%	25%	10% Loss	Discount > Markup → Loss
25%	20%	0% (Break-even)	Discount < Markup but lower

Analysis of 20% Markup & 20% Discount

```
CP = 100
MP = 100 + 20% of 100 = 120
Discount = 20% of 120 = 24
SP = 120 - 24 = 96
Final = 96 - 100 = -4% Loss

Why? 20% of 100 ≠ 20% of 120
20% of 100 = 20 (what you added)
20% of 120 = 24 (what you subtracted)
You subtracted MORE than you added!
```

■ *Analysis:*
When Markup% = Discount% → ALWAYS A LOSS!
Because discount is calculated on the larger amount (MP).

7. Analyzing Comparative Shopping

Problem: Two shops offer different discounts.

	Shop A	Shop B
Offer	8% discount on all items	Rs 100 off on purchases over Rs 1000
MP of shoes	Rs 1500	Rs 1500

Shop	Discount Amount	Final Price
A	8% of 1500 = 120	1500 – 120 = 1380
B	Rs 100	1500 – 100 = 1400

■ *Analysis:*
Shop A is better by Rs 20, because 8% of 1500 = 120 > 100.

```
If MP = 1200:
Shop A: 8% of 1200 = 96
Shop B: Rs 100 off
Shop B is better by Rs 4

When are they equal?
8% of MP = 100 → 0.08 × MP = 100 → MP = 1250

Below Rs 1250 → Shop B better
Above Rs 1250 → Shop A better
```

8. Comparing Different Selling Scenarios

Scenario 1: Same Profit, Different CP

Seller	CP	SP	Profit	Profit %
A	800	900	100	12.5%
B	2500	2600	100	4%

- Same absolute profit (Rs 100)
- Different profit percentage
- A has better return on investment

Scenario 2: Same Profit %, Different CP

Seller	CP	Profit %	Profit	SP
A	800	20%	160	960
B	2500	20%	500	3000

- Same profit percentage (20%)
- Different absolute profit
- B earns more because of higher investment

■ *Analysis:*
Profit% = Efficiency of investment. Absolute Profit = Actual money earned.
Both are important for different reasons!

9. Analyzing Commission Relationships

Commission %	Value	Commission	Net Amount	Seller's Share
0%	1,000,000	0	1,000,000	100%
5%	1,000,000	50,000	950,000	95%
10%	1,000,000	100,000	900,000	90%
15%	1,000,000	150,000	850,000	85%

■ *Analysis:*
Commission% = 5% means the broker gets 5% of the sale, and the seller receives 95%.
The relationship is complementary: Commission% + Seller's Share% = 100%.

Finding Commission from Net Amount

Problem	Analysis	Solution
Broker gets Rs 30,000, Seller gets Rs 570,000	Value = 30,000 + 570,000 = 600,000	Commission% = 30,000/600,000 × 100 = 5%

10. Analyzing Discount vs Commission

Feature	Discount	Commission
Calculated on	Marked Price	Transaction Value
Who benefits	Customer	Broker
Reduces	Price customer pays	Amount seller receives
Purpose	Attract buyers	Pay for service

```
Item: MP = 100,000

Option A: 10% Discount
Discount = 10,000; Customer pays = 90,000

Option B: 10% Commission
Commission = 10,000; Broker gets = 10,000; Seller receives = 90,000
```

■ *Analysis:*
Same amount, but different stakeholders benefit!

11. Analyzing Percentage Word Problems

Word Problem 1: Finding Unknown CP

A vendor buys a stock of mangoes and sells all of them at Rs 30 each, earning a profit of 20%. What was the cost price per mango?

```
SP per mango = 30; Profit% = 20%; Need to find CP

Let CP = x
Profit = 20% of x = 0.2x
SP = CP + Profit = x + 0.2x = 1.2x
1.2x = 30
x = 25

CP = Rs 25
Verification: 20% of 25 = 5 → SP = 30 ✓
```

Word Problem 2: Multiple Selling Prices

Vendor buys 60 kg at Rs 50/kg. Sells 20 kg at Rs 70, 15 kg at Rs 60, 10 kg at Rs 50, 5 kg at Rs 40. Find profit/loss %.

```
Total CP = 60 × 50 = 3000
Total SP = 20×70 + 15×60 + 10×50 + 5×40
Total SP = 1400 + 900 + 500 + 200 = 3000

Conclusion: No profit, no loss (break-even)
```

- Some parts sold at profit (Rs 70, Rs 60)
- Some sold at loss (Rs 40)
- Overall, they balance out

12. Analyzing the Relationship: Percentage, Fraction, Ratio

Percentage	Fraction	Ratio	Meaning
50%	1/2	1:2	Half
25%	1/4	1:4	Quarter
75%	3/4	3:4	Three quarters
20%	1/5	1:5	One fifth
33.33%	1/3	1:3	One third

```
50% = 50/100 = 1/2 = 1:2
25% = 25/100 = 1/4 = 1:4
75% = 75/100 = 3/4 = 3:4

All are different ways of expressing the same thing!
```

13. Analyzing the Earnings Pattern

Salary	Increase %	New Salary	Increase Amount
10,000	5%	10,500	500
10,000	10%	11,000	1,000
10,000	15%	11,500	1,500
10,000	20%	12,000	2,000

- Each 5% increase adds Rs 500
- Percentage and actual increase are proportional
- 10% of 10,000 = 1,000 ✓

```
If salary = 20,000 and increase = 10%
Increase = 2,000; New Salary = 22,000

Pattern: For 5% of 20,000 = 1,000
For 10% of 20,000 = 2,000
```

14. Analysis Checklist

When analyzing a percentage problem:

- What is the base for the percentage? (CP, MP, Value, or something else?)
- Is the percentage being added or subtracted?
- What is the relationship between components?

When comparing transactions:

- What are the absolute values?
- What are the percentage values?
- Which is more profitable/efficient?

When analyzing multi-step problems:

- Is each percentage calculated on the correct base?
- Can percentages be added directly?
- Should I calculate step by step?

When analyzing relationships:

- What is the CP → MP → SP relationship?
- What is the Markup% → Discount% relationship?
- What is the Commission% → Net Amount relationship?

When analyzing word problems:

- What is given and what is asked?
- What operation is needed?
- Does the answer make sense?

15. Self-Test — Analysis Questions

No.	Question	Answer & Justification
1	Why is profit% based on CP, not SP?	Profit% shows return on investment (what you spent)
2	Why is discount% based on MP, not CP?	Discount is a reduction from the marked price
3	What happens when markup% = discount%?	There is always a loss (4% if 20% markup & 20% discount)
4	Why are profit percentages not additive?	Each is calculated on a different CP (base changes)
5	Which is better: 10% profit on CP=1000 or 10% profit on CP=2000?	Both have same profit%, but second has higher absolute profit
6	Why does 20% markup + 20% discount = 4% loss?	Discount is on larger amount (MP), markup on smaller amount (CP)
7	What is the relationship between commission% and seller's share%?	They add up to 100%
8	How does discount percentage compare to actual discount amount?	Higher MP means larger actual discount for same %
9	Why is percentage useful for comparing?	It standardizes values to "per 100"
10	What's the difference between 10% of 100 and 10% of 1000?	Same percentage, but different actual values (10 vs 100)

16. Key Analytical Insights

- ★ **PROFIT% AND LOSS% ALWAYS USE CP AS BASE.**
- ★ **DISCOUNT% ALWAYS USES MP AS BASE.**
- ★ **COMMISSION% ALWAYS USES TRANSACTION VALUE AS BASE.**
- ★ **20% MARKUP + 20% DISCOUNT = 4% LOSS (NOT BREAK-EVEN!)**
- ★ **PERCENTAGES ARE NOT ADDITIVE WHEN BASES CHANGE.**
- ★ **PROFIT% = EFFICIENCY; ABSOLUTE PROFIT = ACTUAL EARNINGS.**
- ★ **HIGHER % DOESN'T ALWAYS MEAN HIGHER ACTUAL VALUE.**
- ★ **COMPARE TRANSACTIONS USING PERCENTAGES, NOT JUST ABSOLUTE VALUES.**
- ★ **PERCENTAGE, FRACTION, AND RATIO ARE DIFFERENT FORMS OF THE SAME THING.**
- ★ **ANALYZING HELPS YOU UNDERSTAND "WHY" NOT JUST "HOW".**

Revision Tip: When analyzing percentage problems, always ask: "What is the base?" and "What is the relationship between these components?" Understanding the "why" helps you solve ANY problem!